

Blood Bank Management System (BBMS)

Software Requirements Specification (SRS)

Software Engineering Mini-Project

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, and security requirements for the Blood Bank Management System (BBMS), a web-based application developed to manage blood donation, testing, inventory, and hospital requisition workflows. This document is prepared as Deliverable Part 1 of a university Software Engineering mini-project and follows an IEEE Std 830-style structure.

1.2 Scope

BBMS is a standalone web application that supports the following in-scope capabilities:

- Donor registration, authentication, profile management, and appointment scheduling
- Medical eligibility screening prior to donation
- Blood collection recording, unique blood-unit (Bag) identification, and blood-group management
- Serology/infection testing, quarantine, and safe-disposal workflows
- Blood component separation and expiry management using First-Expiry-First-Out (FEFO)
- Real-time blood inventory tracking across blood bank locations
- Hospital registration, blood requisition (including emergency/STAT requests), compatibility cross-matching, allocation, dispatch, and delivery confirmation
- Role-based administration, audit logging, reporting/analytics, and public blood-availability lookup

The following are explicitly out of scope for this mini-project:

- Integration with national or regional blood-bank registries or government health systems
- Payment processing or billing of any kind
- Direct interfacing with physical laboratory testing equipment (results are entered manually by staff)
- Native mobile applications (the system is delivered as a responsive web application)
- Multi-language localization

1.3 Audience

This document is intended for the course instructor/evaluator, the student development team responsible for design and implementation, and any future maintainer of the BBMS codebase.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
BBMS	Blood Bank Management System
SRS	Software Requirements Specification
FEFO	First-Expiry-First-Out
RBC	Red Blood Cells
RBAC	Role-Based Access Control
API	Application Programming Interface
REST	Representational State Transfer
HTTPS/TLS	Hypertext Transfer Protocol Secure / Transport Layer Security
ORM	Object-Relational Mapping
UI	User Interface
UML	Unified Modeling Language
XSS	Cross-Site Scripting
SQLi	SQL Injection
STAT	Immediate / emergency priority request
Bag ID	Unique identifier assigned to a collected blood unit
WCAG	Web Content Accessibility Guidelines

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications — used as a structural reference for this document.
- General publicly available domain background on blood-bank operating workflows (donor screening, testing, FEFO inventory practice) — used only to inform realistic system behaviour, not as a certified regulatory source.

1.6 Document Overview

Section 2 describes the product at a high level. Section 3 defines external interfaces. Section 4 lists functional requirements by module. Section 5 lists non-functional requirements. Section 6 defines security objectives and requirements. Section 7 presents UML use-case diagrams. Section 8 provides a requirements traceability matrix linking requirements to modules and test cases defined in the companion Test Plan.

2. Overall Description

2.1 Product Perspective

BBMS is a new, self-contained web application developed for academic demonstration purposes. It is not a replacement for, nor does it integrate with, any existing legacy blood-bank system. The system is architected to resemble a realistic blood-bank management workflow (donor → collection → testing → inventory → requisition → dispatch) at a scale appropriate for a university mini-project.

2.2 Major Product Functions

- Donor management (registration, profile, eligibility, appointments)
- Blood collection, testing, quarantine, and component separation
- Inventory management with expiry (FEFO) handling
- Hospital management and blood requisition/dispatch workflow
- Compatibility cross-matching for safe allocation
- Role-based administration and audit logging
- Reporting and analytics
- Public blood-availability lookup

2.3 User Classes and Characteristics

User Class	Description
Donor	A registered individual who books appointments, donates blood, and views personal donation history. Assumed to have basic computer literacy.
Hospital User	A representative of a registered hospital who submits blood requisitions, tracks their status, and confirms delivery. Assumed to have basic computer literacy and domain familiarity with blood-component terminology.
Blood Bank Staff	Laboratory/collection personnel who record collections, perform and log serology testing, manage quarantine and component separation, manage inventory, approve requisitions, and perform allocation and dispatch.
Administrator	Manages user accounts and roles, approves hospital registrations, views the complete audit log, and generates system-wide reports.
Public / Guest	An unauthenticated visitor who may view the public blood-availability page only.

2.4 Operating Environment

- Server: runs on a standard PC, laptop, or single virtual machine; no specialized hardware required.
- Backend runtime: Python 3.10+ (Django) or Node.js 18+ (Express), per the implementing team's technology choice.
- Database: SQLite (file-based, embedded), appropriate for the mini-project's scale.
- Client: any device with a modern, JavaScript-enabled web browser and internet/LAN connectivity to the server.

2.5 Design and Implementation Constraints

- The system must be implemented using the specified stack: Python (Django) or Node.js backend, JavaScript/HTML/CSS frontend, SQLite database.
- SQLite has limited concurrent-write support; the system is designed for the concurrency levels of a university mini-project demonstration, not a production multi-hospital network.

- The project must be delivered by the fixed Deliverable Part 1 deadline of 28 September 2026, which constrains the depth of implementation (though not the completeness of this specification).
- The system is an academic prototype and is not certified for real clinical or regulatory use.

2.6 Assumptions and Dependencies

- Users have access to a device with internet or LAN connectivity and a modern web browser.
- Donors self-report accurate information during registration and the eligibility questionnaire.
- Physical laboratory testing (serology) is performed by blood bank staff using equipment outside the system; staff enter results into BBMS manually.
- All system components run with synchronized clocks (server time is authoritative for timestamps, expiry calculations, and audit logs).

3. External Interface Requirements

3.1 User Interfaces

BBMS provides a responsive, browser-based web interface with distinct dashboards for each user class (Donor, Hospital, Staff, Administrator). Navigation menus and available actions are rendered according to the authenticated user's role. All data-entry forms perform client-side validation in addition to server-side validation.

3.2 Hardware Interfaces

BBMS does not interface directly with specialized hardware. Client devices require only standard input (keyboard/touch) and display capability sufficient to render a modern web browser. No barcode scanners, lab analyzers, or other devices are integrated in this mini-project scope.

3.3 Software Interfaces

- Backend web framework: Django (Python) or Express (Node.js).
- Data access: an Object-Relational Mapping (ORM) layer (e.g., Django ORM or Sequelize/Prisma) over a SQLite database file.
- Outbound notification interface: SMTP for sending email notifications (registration confirmation, appointment updates, password reset, emergency-requisition alerts).

3.4 Communication Interfaces

- All client-server communication shall use HTTPS.
- The frontend communicates with the backend via a RESTful JSON API (see the companion Software Architecture & Design Specification for endpoint details).
- Outbound email notifications are sent via SMTP.

4. Functional Requirements

Each functional requirement is uniquely identified, atomic, measurable, testable, and stated using “shall.” Requirements are grouped by module below.

4.1 Donor Registration and Login

ID	Requirement Description	Priority
BBMS-F-001	The system shall allow a new donor to register by providing full name, date of birth, gender, contact number, email address, and a password.	High
BBMS-F-002	The system shall validate that the donor is at least 18 years old at the time of registration, calculated from the provided date of birth.	High
BBMS-F-003	The system shall enforce a unique email address per donor account and shall reject registration attempts using an email already in use.	High
BBMS-F-004	The system shall authenticate a registered donor using email and password before granting access to donor-only functions.	High
BBMS-F-005	The system shall allow a donor to reset a forgotten password via a time-limited, single-use reset link sent to the registered email address.	Medium

4.2 Donor Profile

ID	Requirement Description	Priority
BBMS-F-006	The system shall allow a logged-in donor to view and update personal profile details, including contact number, address, and blood group.	Medium
BBMS-F-007	The system shall display the donor's donation history, including date, location, and Bag ID of each past donation.	Medium
BBMS-F-008	The system shall display the donor's next eligible donation date, calculated as a configurable minimum interval (default 90 days) after the last successful donation.	Medium

4.3 Medical Eligibility Questionnaire

ID	Requirement Description	Priority
BBMS-F-009	The system shall present a standardized medical eligibility questionnaire to a donor before confirming an appointment booking.	High
BBMS-F-010	The system shall automatically mark a donor as temporarily ineligible if any questionnaire answer matches a configured disqualifying condition, and shall display the reason to the donor.	High
BBMS-F-011	The system shall record the questionnaire responses and eligibility outcome against the associated appointment for audit purposes.	Medium

4.4 Donation Appointment Booking, Rescheduling, and Cancellation

ID	Requirement Description	Priority
BBMS-F-012	The system shall allow an eligible donor to book a donation appointment by selecting an available date, time slot, and blood bank location.	High
BBMS-F-013	The system shall prevent booking of a time slot that has reached its configured donor capacity.	High
BBMS-F-014	The system shall allow a donor to reschedule an existing appointment to another available slot up to 24 hours before the scheduled time.	Medium

ID	Requirement Description	Priority
BBMS-F-015	The system shall allow a donor to cancel an existing appointment and shall release the reserved slot capacity upon cancellation.	Medium
BBMS-F-016	The system shall send a confirmation notification (on-screen and/or email) when an appointment is booked, rescheduled, or cancelled.	Low

4.5 Blood Collection

ID	Requirement Description	Priority
BBMS-F-017	The system shall allow blood bank staff to record the start and completion of a blood collection against a confirmed donor appointment.	High
BBMS-F-018	The system shall require blood bank staff to record the collected volume (in millilitres) and collection date/time for each donation.	High
BBMS-F-019	The system shall update the donor's last-donation date and donation history immediately upon confirming a successful collection.	High

4.6 Unique Blood Unit / Bag ID

ID	Requirement Description	Priority
BBMS-F-020	The system shall automatically generate a unique Bag ID for every collected blood unit at the time collection is recorded.	High
BBMS-F-021	The system shall prevent the creation of two blood units with the same Bag ID.	High

4.7 Blood Group Management

ID	Requirement Description	Priority
BBMS-F-022	The system shall support the eight standard ABO/Rh blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-) for donor and blood-unit records.	High
BBMS-F-023	The system shall allow blood bank staff to record or correct the laboratory-confirmed blood group of a blood unit, distinct from the donor's self-reported blood group.	High

4.8 Serology / Infection Testing

ID	Requirement Description	Priority
BBMS-F-024	The system shall require every collected blood unit to undergo recorded serology testing for HIV, Hepatitis B, Hepatitis C, and Syphilis before it can be marked available for use.	High
BBMS-F-025	The system shall allow blood bank staff to record a Pass or Fail result for each serology test performed on a blood unit.	High
BBMS-F-026	The system shall set the status of a blood unit to "Quarantined" by default upon collection and shall not permit a change to "Available" status until all required test results are recorded as Pass.	High

ID	Requirement Description	Priority
BBMS-F-027	The system shall set the status of a blood unit to "Discarded" if any serology test result is recorded as Fail, and shall prevent such units from being allocated or dispatched.	High

4.9 Quarantine of Unsafe Blood

ID	Requirement Description	Priority
BBMS-F-028	The system shall exclude blood units with a "Quarantined" or "Discarded" status from inventory availability counts and hospital search results.	High
BBMS-F-029	The system shall log the identity of the staff member and timestamp whenever a blood unit's status changes to or from "Quarantined."	High
BBMS-F-030	The system shall allow authorized blood bank staff to view a list of all currently quarantined blood units together with the elapsed quarantine duration.	Medium

4.10 Blood Component Separation

ID	Requirement Description	Priority
BBMS-F-031	The system shall allow blood bank staff to record the separation of a whole-blood unit into components (e.g., Red Blood Cells, Plasma, Platelets), generating a distinct inventory record for each resulting component.	Medium
BBMS-F-032	The system shall retain a reference from each separated component record to its originating whole-blood unit Bag ID for traceability.	Medium

4.11 Blood Inventory Management

ID	Requirement Description	Priority
BBMS-F-033	The system shall maintain a real-time count of available blood units per blood group and component type at each blood bank location.	High
BBMS-F-034	The system shall allow blood bank staff to view the complete inventory, filterable by blood group, component type, and status.	High
BBMS-F-035	The system shall automatically decrement the available inventory count when a blood unit is allocated to a hospital requisition.	High
BBMS-F-036	The system shall automatically increment the available inventory count when a new blood unit is marked "Available" after passing serology testing.	High

4.12 Expiry Handling and FEFO

ID	Requirement Description	Priority
BBMS-F-037	The system shall calculate and store an expiry date for each blood unit based on its component type at the time it is marked "Available" (e.g., 42 days for whole blood/RBC, 5 days for platelets, 1 year for frozen plasma).	High

ID	Requirement Description	Priority
BBMS-F-038	The system shall apply First-Expiry-First-Out (FEFO) ordering when suggesting or automatically selecting blood units for allocation to a requisition.	High
BBMS-F-039	The system shall automatically flag a blood unit's status as "Expired" once its expiry date has passed and shall exclude expired units from availability counts and allocation.	High

4.13 Hospital Registration and Login

ID	Requirement Description	Priority
BBMS-F-040	The system shall allow a hospital to register by providing hospital name, registration/license number, address, contact person, and contact details, subject to administrator approval before activation.	High
BBMS-F-041	The system shall authenticate a registered hospital user using hospital-specific login credentials before granting access to requisition functions.	High
BBMS-F-042	The system shall prevent an unapproved hospital account from submitting blood requisitions.	High

4.14 Blood Requisition

ID	Requirement Description	Priority
BBMS-F-043	The system shall allow an authenticated hospital user to submit a blood requisition specifying blood group, component type, quantity of units, and required-by date.	High
BBMS-F-044	The system shall assign each submitted requisition a unique Requisition ID and an initial status of "Pending."	High
BBMS-F-045	The system shall allow blood bank staff to approve, partially fulfil, or reject a pending requisition, recording a reason for any rejection.	High
BBMS-F-046	The system shall allow a hospital user to view the current status and history of their submitted requisitions.	Medium

4.15 Emergency / STAT Requests

ID	Requirement Description	Priority
BBMS-F-047	The system shall allow a hospital user to mark a requisition as "Emergency/STAT," which shall place it at the top of the blood bank staff's pending-requisition queue.	High
BBMS-F-048	The system shall notify available blood bank staff, via on-screen alert and/or email, immediately upon submission of an Emergency/STAT requisition.	High

4.16 Blood Compatibility and Cross-Matching

ID	Requirement Description	Priority
BBMS-F-049	The system shall determine and display ABO/Rh-compatible blood groups for a requested blood group before allocation, based on standard transfusion-compatibility rules.	High
BBMS-F-050	The system shall prevent blood bank staff from allocating a blood unit whose blood group is not compatible with the requisition's requested blood group, unless an authorized staff member records an explicit override with justification.	High
BBMS-F-051	The system shall record the cross-matching outcome (Compatible / Incompatible / Overridden) against each allocation.	High

4.17 Allocation and Dispatch

ID	Requirement Description	Priority
BBMS-F-052	The system shall allow blood bank staff to allocate one or more specific blood units, by Bag ID, to an approved requisition.	High
BBMS-F-053	The system shall generate a dispatch record containing the requisition ID, allocated Bag ID(s), dispatching staff identity, and dispatch timestamp.	High
BBMS-F-054	The system shall update a dispatched blood unit's status to "Dispatched" and remove it from available inventory counts.	High
BBMS-F-055	The system shall update the requisition status to "Fulfilled" once all requested units have been dispatched, or to "Partially Fulfilled" if fewer units are dispatched than requested.	High

4.18 Delivery Confirmation

ID	Requirement Description	Priority
BBMS-F-056	The system shall allow a hospital user to confirm receipt of a dispatched blood unit, recording the confirmation timestamp and receiving staff identity.	Medium
BBMS-F-057	The system shall update a blood unit's status to "Delivered" upon hospital confirmation of receipt.	Medium

4.19 Audit Logging

ID	Requirement Description	Priority
BBMS-F-058	The system shall record an audit log entry for every creation, status change, or deletion of a donor, blood unit, requisition, or user account, capturing the acting user, action performed, timestamp, and affected record ID.	High
BBMS-F-059	The system shall restrict access to the complete audit log to users with the Administrator role, and shall render audit log entries read-only (non-editable, non-deletable) to all users.	High

4.20 Admin, User, and Role Management

ID	Requirement Description	Priority
BBMS-F-060	The system shall allow an Administrator to create, view, deactivate, and reactivate user accounts for all roles (Donor, Hospital, Blood Bank Staff, Administrator).	High
BBMS-F-061	The system shall enforce Role-Based Access Control (RBAC) such that each system function is accessible only to the user roles authorized for it, as defined in the system's role-permission matrix.	High
BBMS-F-062	The system shall allow an Administrator to approve or reject pending hospital registration requests.	High
BBMS-F-063	The system shall prevent a deactivated user account from authenticating into the system.	High

4.21 Reports and Analytics

ID	Requirement Description	Priority
BBMS-F-064	The system shall generate a summary report of current inventory levels by blood group and component type, exportable in CSV format.	Medium
BBMS-F-065	The system shall generate a donation activity report for a specified date range, showing total donations, units collected, and units discarded.	Medium
BBMS-F-066	The system shall generate a requisition fulfillment report showing requisitions received, fulfilled, partially fulfilled, and rejected for a specified date range.	Medium

4.22 Public Blood Availability

ID	Requirement Description	Priority
BBMS-F-067	The system shall provide a public-facing page displaying current blood availability by blood group and blood bank location, accessible without authentication.	Medium
BBMS-F-068	The system shall exclude donor-identifying and hospital-identifying information from the public blood availability page.	High

5. Non-Functional Requirements

5.1 Performance

ID	Requirement Description	Priority
BBMS-NF-001	The system shall return results for a blood-availability search within 2 seconds under normal load (up to 50 concurrent users).	High
BBMS-NF-002	The system shall complete a donor appointment-booking transaction, from request to confirmation, within 3 seconds.	Medium
BBMS-NF-003	The system shall support at least 50 concurrent authenticated users without response times exceeding the limits stated in BBMS-NF-001.	Medium

5.2 Reliability and Availability

ID	Requirement Description	Priority
BBMS-NF-004	The system shall be available for use at least 99% of scheduled operating hours, excluding planned maintenance windows.	Medium
BBMS-NF-005	The system shall retain all committed transaction data (donations, requisitions, dispatches) across an application restart, with no data loss for committed transactions.	High
BBMS-NF-006	The system shall log unhandled application errors, with timestamp and stack trace, to a server-side error log for diagnosis.	Medium

5.3 Usability and Accessibility

ID	Requirement Description	Priority
BBMS-NF-007	A first-time user shall be able to complete donor registration without external assistance in under 5 minutes, as verified by a usability walkthrough.	Medium
BBMS-NF-008	The system's web interface shall conform to WCAG 2.1 Level AA colour-contrast and keyboard-navigation guidelines for all primary user workflows.	Medium
BBMS-NF-009	The system shall display clear, human-readable error messages for all form-validation failures, identifying the specific field and issue.	High

5.4 Maintainability

ID	Requirement Description	Priority
BBMS-NF-010	The system's backend shall follow a layered/modular code structure such that a change to one module (e.g., inventory) does not require modification of unrelated modules (e.g., reporting).	Medium
BBMS-NF-011	The system shall maintain a minimum of 60% automated unit-test coverage for backend business-logic modules.	Low

5.5 Portability

ID	Requirement Description	Priority
BBMS-NF-012	The system's web client shall render correctly on the latest two stable versions of Chrome, Firefox, and Edge.	Medium
BBMS-NF-013	The system's backend shall be deployable on any host operating system supporting Python 3.10+ or Node.js 18+ without code modification.	Low

5.6 Scalability

ID	Requirement Description	Priority
BBMS-NF-014	The system's database schema shall support migration from SQLite to a multi-user RDBMS (e.g., PostgreSQL) without change to the application's data-access interfaces.	Low
BBMS-NF-015	The system's architecture shall allow the addition of a new blood bank location without modification of the existing donor, hospital, or inventory modules.	Low

6. Security

6.1 Security Objectives

- Protect the confidentiality of donor personal and medical information from unauthorized access or disclosure.
- Ensure that only authenticated and authorized users can perform actions appropriate to their assigned role.
- Preserve the integrity of blood-unit status and testing records to prevent unsafe blood from being allocated or dispatched.
- Maintain accountability through a non-repudiable audit trail of all security-relevant and data-changing actions.

6.2 Security Requirements

ID	Requirement Description	Covers
BBMS-SR-001	The system shall require all users (Donor, Hospital, Blood Bank Staff, Administrator) to authenticate with a unique email/username and password before accessing any role-specific function.	Authentication
BBMS-SR-002	The system shall store user passwords using a salted, one-way hashing algorithm (e.g., bcrypt) and shall never store or transmit passwords in plaintext.	Password hashing
BBMS-SR-003	The system shall enforce role-based access control (RBAC) on the server side for every API endpoint, independent of client-side UI restrictions, and shall deny unauthorized requests with an HTTP 403 response.	RBAC / authorization
BBMS-SR-004	The system shall use parameterized queries or an Object-Relational Mapping (ORM) layer for all database operations and shall not construct SQL statements by concatenating raw user input.	SQL injection prevention
BBMS-SR-005	The system shall validate and sanitize all user-supplied input on both client and server sides, and shall encode output rendered in the browser to prevent stored and reflected Cross-Site Scripting (XSS).	Input validation / XSS
BBMS-SR-006	The system shall issue session tokens as HttpOnly, Secure cookies with a defined expiry, and shall invalidate a session upon logout or after 30 minutes of inactivity.	Session / cookie security
BBMS-SR-007	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and shall rate-limit login requests to no more than 10 per minute per source IP address.	Brute-force / rate limiting

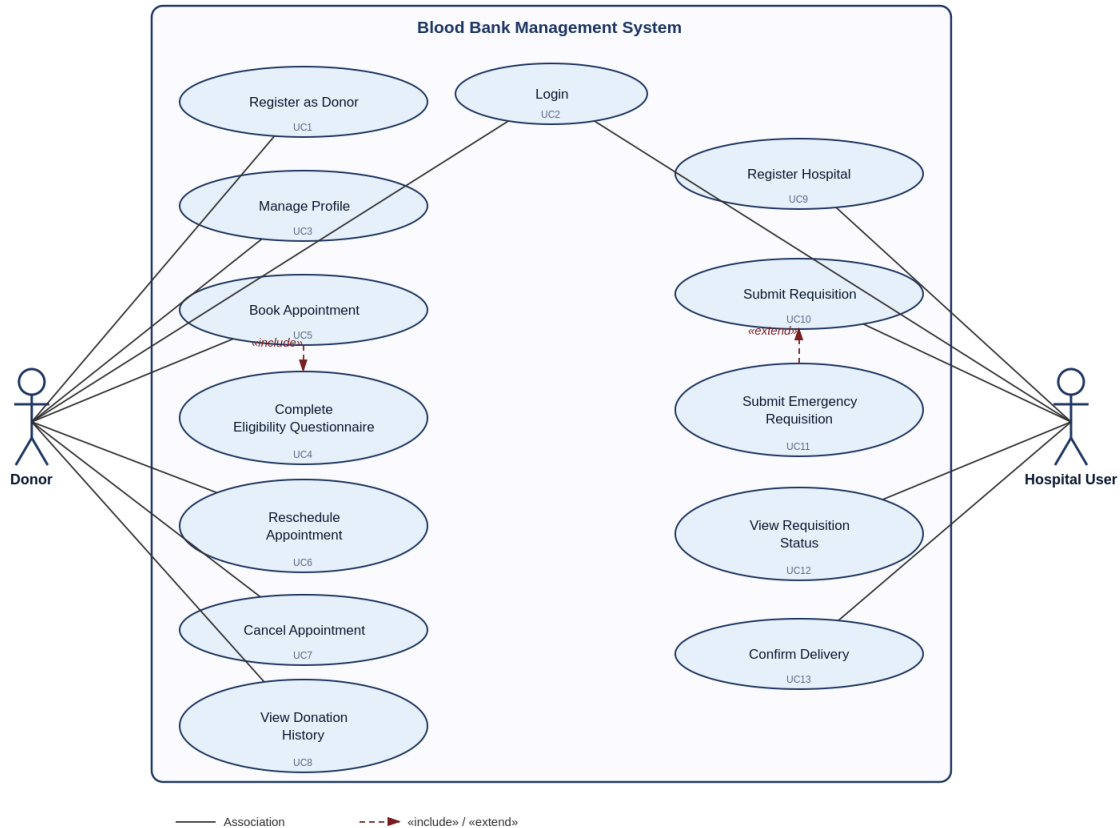
ID	Requirement Description	Covers
BBMS-SR-008	The system shall transmit all client-server communication over HTTPS/TLS and shall record a tamper-evident audit trail (per BBMS-F-058) of security-relevant events, including login attempts, account lockouts, and role changes.	Confidentiality, integrity, audit

7. UML Use-Case Diagrams

The system boundary is the Blood Bank Management System (BBMS). Equivalent Mermaid source is included after each diagram so it can also be pasted into a Markdown/GitHub viewer.

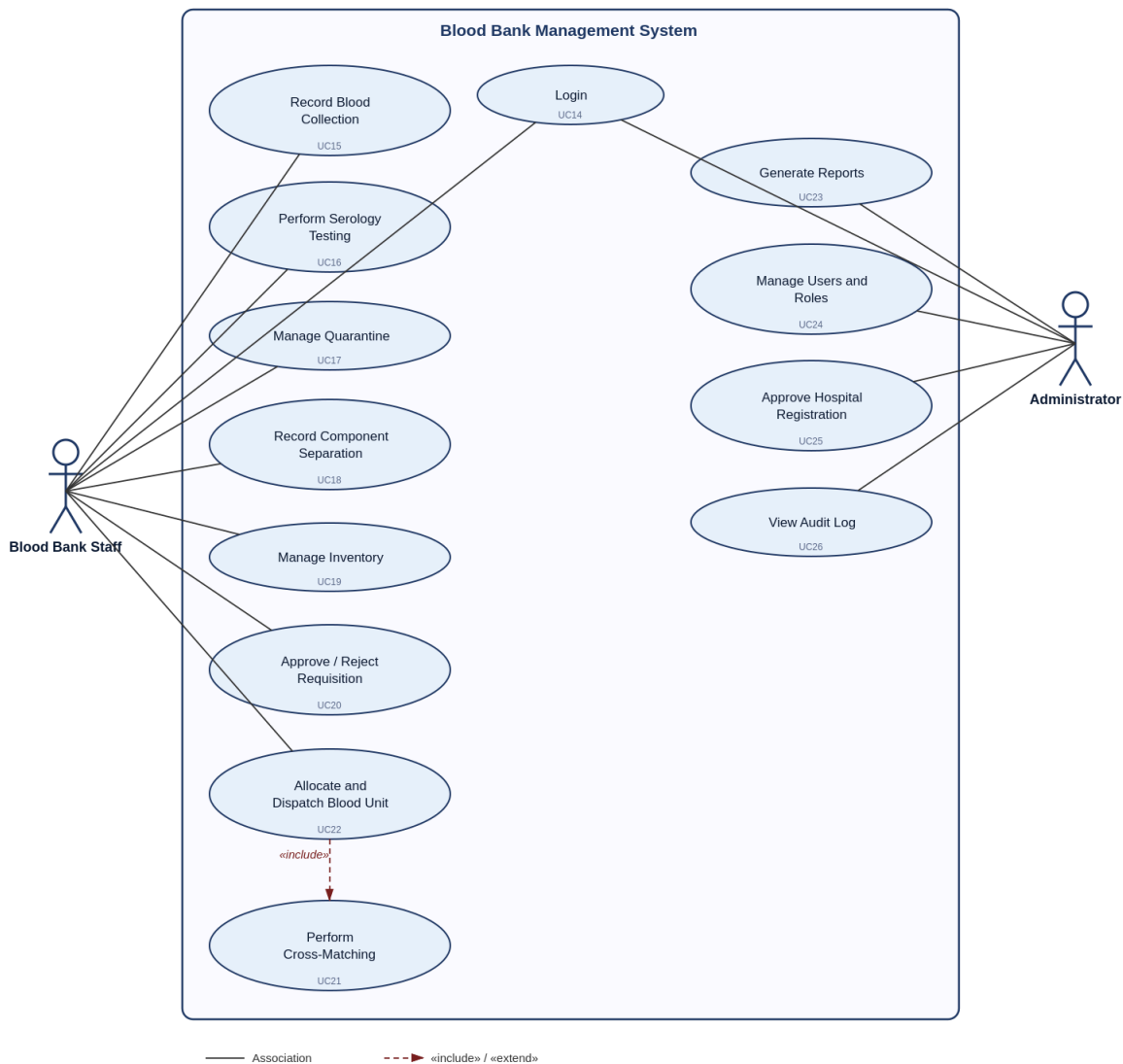
7.1 Use-Case Diagram — Donor and Hospital Workflows

Actors: Donor, Hospital User. “Book Appointment” includes “Complete Eligibility Questionnaire” (BBMS-F-009). “Submit Emergency Requisition” extends “Submit Requisition” (BBMS-F-047).



7.2 Use-Case Diagram — Staff and Administrator Workflows

Actors: Blood Bank Staff, Administrator. “Allocate and Dispatch Blood Unit” includes “Perform Cross-Matching” (BBMS-F-049).



8. Requirements Traceability Matrix

This matrix maps each requirement group to its module and to the Test Case ID(s) defined in the companion Test Plan. Status values follow the Test Plan's convention ("Not Run", "Pass", "Fail", "Blocked") and are set to "Not Run" pending execution.

Requirement ID(s)	Module / Feature	Test Case ID(s)	Status
BBMS-F-001 – F-005	Donor Registration / Login	TC-AUTH-01, TC-AUTH-02	Not Run
BBMS-F-006 – F-008	Donor Profile	TC-DNR-01	Not Run
BBMS-F-009 – F-011	Medical Eligibility Questionnaire	TC-ELG-01	Not Run
BBMS-F-012 – F-016	Appointment Booking / Reschedule / Cancel	TC-APT-01, TC-APT-02	Not Run

Requirement ID(s)	Module / Feature	Test Case ID(s)	Status
BBMS-F-017 – F-019	Blood Collection	TC-COL-01	Not Run
BBMS-F-020 – F-021	Unique Blood Unit / Bag ID	TC-BAG-01	Not Run
BBMS-F-022 – F-023	Blood Group Management	TC-BAG-02	Not Run
BBMS-F-024 – F-027	Serology / Infection Testing	TC-SER-01, TC-SER-02	Not Run
BBMS-F-028 – F-030	Quarantine of Unsafe Blood	TC-QNT-01	Not Run
BBMS-F-031 – F-032	Blood Component Separation	TC-CMP-01	Not Run
BBMS-F-033 – F-036	Blood Inventory Management	TC-INV-01, TC-INV-02	Not Run
BBMS-F-037 – F-039	Expiry Handling / FEFO	TC-EXP-01	Not Run
BBMS-F-040 – F-042	Hospital Registration / Login	TC-AUTH-03	Not Run
BBMS-F-043 – F-046	Blood Requisition	TC-REQ-01, TC-REQ-02	Not Run
BBMS-F-047 – F-048	Emergency / STAT Requests	TC-REQ-03	Not Run
BBMS-F-049 – F-051	Compatibility / Cross-Matching	TC-MATCH-01	Not Run
BBMS-F-052 – F-055	Allocation and Dispatch	TC-DISP-01, TC-DISP-02	Not Run
BBMS-F-056 – F-057	Delivery Confirmation	TC-DEL-01	Not Run
BBMS-F-058 – F-059	Audit Logging	TC-AUD-01	Not Run
BBMS-F-060 – F-063	Admin / User / Role Management	TC-ADM-01, TC-SEC-03	Not Run
BBMS-F-064 – F-066	Reports and Analytics	TC-RPT-01	Not Run
BBMS-F-067 – F-068	Public Blood Availability	TC-PUB-01	Not Run
BBMS-NF-001 – NF-003	Performance	TC-NF-01	Not Run
BBMS-NF-004 – NF-006	Reliability / Availability	TC-NF-02	Not Run
BBMS-NF-007 – NF-009	Usability / Accessibility	TC-NF-03	Not Run
BBMS-NF-010 – NF-011	Maintainability	TC-NF-04	Not Run
BBMS-NF-012 – NF-013	Portability	TC-NF-05	Not Run
BBMS-NF-014 – NF-015	Scalability	TC-NF-06	Not Run
BBMS-SR-001	Authentication	TC-SEC-01	Not Run
BBMS-SR-002	Password Hashing	TC-SEC-02	Not Run
BBMS-SR-003	RBAC / Authorization	TC-SEC-03	Not Run
BBMS-SR-004	SQL Injection Prevention	TC-SEC-04	Not Run
BBMS-SR-005	Input Validation / XSS	TC-SEC-05	Not Run
BBMS-SR-006	Session / Cookie Security	TC-SEC-06	Not Run

Requirement ID(s)	Module / Feature	Test Case ID(s)	Status
BBMS-SR-007	Brute-Force / Rate Limiting	TC-SEC-07	Not Run
BBMS-SR-008	Audit / Confidentiality / Integrity	TC-SEC-08, TC-AUD-01	Not Run